Candela Lucia Szischik | Curriculum Vitae

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Born 1st December 1995. Buenos Aires, Argentina

RESEARCH PROFILE

My research lies at the interface between physics and biology, with a focus on gene regulatory networks and microRNA-mediated regulation. I have a strong background in nonlinear dynamics and signal processing in biological systems, and experience integrating theoretical modeling with experimental approaches, including molecular biology techniques. I am interested in information processing and decision-making in cellular systems, as well as applying physics-based approaches to problems in the physics of living systems.

EDUCATION

PhD. in Physics. 2021 – present. Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. Instituto de Fisiología, Biología Molecular y Neurociencias (IFIBYNE, CONICET UBA).
Advisor: Dra. Alejandra C. Ventura. **Co-advisor:** Dr. Manuel de la Mata. **Thesis:** *Study of gene regulatory networks with interacting mRNAs and microRNAs, using a theoretical-experimental approach from physics.*
Courses: *Biological Physics, Statistical Methods for Experimental Physics, New Advances in Thermodynamics and Statistical Physics, New Mathematical and Computational Methods for Science and Engineering, Dynamic Systems and Artificial Intelligence Applied to Data Modeling.*

Degree in Physics (Bachelor and MSc. equivalence) 2014 – 2021. Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. Average grade: 9.32. Thesis tutor: Dra. Marina Inchaussandague. **Thesis**: *Electromagnetic response in corrugated multilayer systems: a study of the mirror effect in the butterfly Dione Vanillae.* Grade: 10/10.

RESEARCH EXPERIENCE

PhD Researcher, Instituto de Fisiología, Biología Molecular y Neurociencias (2021- present). Area: biophysics, systems biology, cell signalling and gene regulatory networks.

Developed theoretical models of gene regulatory networks involving microRNA–mRNA interactions. Studied dynamical properties of signaling systems using tools from nonlinear dynamics and statistical physics. Combined computational modeling with experimental approaches in molecular biology. Collaborated in interdisciplinary projects in systems biology and biophysics.

Collaborator as a fellowship researcher in the following grants:

- **ANPCyT (PICT2019) 2021-2024:** Dynamic restructuring of feedback systems in cell signaling: multidisciplinary approach. Prof. Alejandra C. Ventura.
- **ANPCyT (PICT2019) 2021-2024:** Systems Biology in signal transduction pathways: transient: cellular memory and irreversible decision making. Prof. Alejandro Colman-Lerner.
- **Visiting scientist at the Statistical Inference and Computational Biology group** at the Italian Institute for Genomic Medicine, Turin, Italy. (2022-2025) Group Leader: Prof. Andrea Pagnani.

Undergraduate student within the research project UBACyT Mod I in Bionanophotonics (2018 – 2019) Advisor: Prof. Diana Skigin. Universidad de Buenos Aires. Area: Applied electromagnetism, nanobiophotonics. Studied electromagnetic properties of biological photonic structures. Developed numerical and theoretical models for optical systems

RESEARCH VISITS

Research visit – Department of Applied Science and Technology (DISAT) Politecnico di Torino, hosted by Prof. Carla Bosia. Turin, Italy, 7–11 July 2025.

Research internship – Italian Institute of Genomic Medicine (IIGM) and Politecnico di Torino. Under the supervision of Prof. Carla Bosia. Experimental setup for post-transcriptional regulation and optogenetics project: plasmid construction, cell culture. Turin, Italy, September-November 2024.

Research stay – IIGM and Politecnico di Torino. Supervised by Prof. Carla Bosia. Preparation of a manuscript on theoretical research of oscillations in post-transcriptional regulation. October 2023.

Research internship – IIGM and Politecnico di Torino. Supervised by Dr. Carla Bosia. Training in molecular biology techniques: cloning (digestion, ligation, restriction mapping, mini/midi-prep, PCR, bacterial transformation), cell culture, and plasmid transfection. October - December 2022.

FELLOWSHIPS AND AWARDS

Poster Award at the European Biophysics Congress (EBSA 2025) June 30th to July 4th 2025, Rome, Italy. *The advantage of periodic over constant signalling in microRNA mediated regulation*, Szischik C, Ferro E, Ventura AC, Bosia C.

Travel Grant – International Union of Pure and Applied Biophysics (IUPAB) Financial support to attend the 15th European Biophysics Congress (EBSA 2025), held in Rome, Italy, from 30 June to 4 July 2025.

Mobility Grant – Graduate Students and Early-Career Researchers Programme (+Viajes). Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. Funding to carry out a

research stay at the Italian Institute of Genomic Medicine and Politecnico di Torino, Turin, Italy, October 2023.

Travel Grant – The Company of Biologists for a collaboration with the Inference and Computational Biology unit, Politecnico of Torino and Italian Institute of Genomic Medicine. Stay from October to December 2022. Experimental research in gene engineering.

Doctoral Fellowship – CONICET (National Scientific and Technical Research Council). Project: Study of gene regulatory networks with interacting mRNAs and microRNAs using a theoretical–experimental physics approach. Supervisor: Dra. Alejandra Ventura. Co-supervisor: Dr. Manuel de la Mata. Host institution: IFIBYNE. Ranked 4th in the national merit list.

Undergraduate Research Fellowship for Scientific Vocation granted by the Consejo Interuniversitario Nacional (Beca EVC-CIN), to develop the research project: *Study of the mirror effect in the wings of the Agraulis Vanillae butterfly* (UBACyT Bionanophotonics project) to accomplish the MSc. Thesis in Physics. 2018

PUBLICATIONS

Ferro E[+], **Szischik CL**[+], Ventura AC, Bosia C. *The advantage of periodic over constant signalling in microRNA-mediated regulation.* Nucleic Acids Res. 2025 Sep 5;53(17) doi: 10.1093/nar/gkaf867.

Ferro E, **Szischik CL**, Cunial M, Ventura AC, De Martino A, Bosia C. *Out-of-Equilibrium ceRNA Crosstalk*. Methods Mol Biol. 2025; 2883:167-193. doi: 10.1007/978-1-0716-4290-0_8.

Szischik CL[+], Reves Szemere J[+], Balderrama R, Sánchez de la Vega C, Ventura AC. *Transient frequency preference responses in cell signaling systems.* npj Syst Biol Appl 10, 86 (2024). https://doi.org/10.1038/s41540-024-00413-w

Szischik CL, Inchaussandague M., and Skigin D., *Electromagnetic response of corrugated multilayer systems inspired by the Dione vanillae butterfly scales*, J. Opt. Soc. Am. A 40, C68-C73 (2023) https://doi.org/10.1364/JOSAA.479677

Dolinko A, Borgmann L, Lutz C, Ronald Curticean E, Wacker I, Vidal MS, **Szischik CL**, Donie Y, Inchaussandague M, Skigin D, Hölscher H, Tubaro P, Barreira A. *Analysis of the optical properties of the silvery spots on the wings of the Gulf Fritillary, Dione vanillae.* Sci Rep 11, 19341 (2021). https://doi.org/10.1038/s41598-021-98237-9

IN PREPARATION

Szischik CL[+], Gándara M[+], Ferro E, Ventura AC, Bosia AC. From threshold-like behavior to microRNA-mediated post-transcriptional decoding. To be sent to the special issue *Dynamic miRNA expression and function* of Transcription, 2026.

[+]*: equal contribution*

CONFERENCE ATTENDANCE AND PRESENTATIONS

International School on Statistical Inference from Biological Data. February 2nd -13th 2026, La Habana, Cuba. The Abdus Salam International Centre for Theoretical Physics (ICTP) and the University La Habana. Invited talk and poster presentation: *Frequency decoding in post-transcriptional regulation.*

LIII Argentinian Biophysics Society (SAB) Annual Meeting. November 26th-28th 2025, Buenos Aires, Argentina. Poster presentation: *Reconceiving microRNAs as post-transcriptional frequency decoders (Szischik C, Ferro E, Ventura A C, Bosia C).*

III IBIOBA Student Workshop: Cell communication and signal transduction. September 17th 2025, Buenos Aires, Argentina. Poster presentation: *Frequency preference responses can emerge from transient dynamics in pulsatile cell signalling events (Szischik C., Reves Szmere J, Balderrama R, De la Vega, C, Ventura A C)*

15th European Biophysics Congress (EBSA 2025). June 30th to July 4th 2025, Rome, Italy. Poster presentation: *The advantage of periodic over constant signalling in microRNA mediated regulation*, Szischik C, Ferro E, Ventura AC, Bosia C. Poster award.

8° Workshop Modelamiento Matemático de Sistemas Biológicos 2025. January 15th- 17th 2025. Virtual meeting.

Dynamics Days Latin America and the Caribbean 2024. December 9th-13th, Buenos Aires, Argentina. Poster presentation: *Frequency preference response can emerge from transient dynamics in pulsatile cell signalling events.*

Nis School 2024: Information, Noise and Physics of Life. The Abdus Salam International Centre for Theoretical Physics, and the Mathematical Institute of the Serbian Academy of Sciences and Arts, University of Niš Faculty of Sciences and Mathematics. June 17th-29th 2024. Nis, Serbia.

Frontiers in Bioscience 4. (IBIOBA - Max Planck Society) Sept 13th – 15th 2023. Buenos Aires, Argentina. Poster presentation titled *Frequency preference in microRNA mediated regulation (Szischik C, Ferro E, Bosia C, Ventura A C).*

Workshop on Signatures of Nonequilibrium Fluctuations in Life. ICTP May 15th - 19th 2023, Trieste, Italy. Poster presentation Frequency response in microRNA-mediated genetic regulation *(Szischik C., Ferro E., Bosia C., Ventura A.C.).*

Encuentro de Matemática y Biología 2023. Departamento de Matemática, FCEN, UBA. April 17th-18th 2023. Buenos Aires, Argentina.

Computational Aspects and Modeling of Biological Information Workshop. December 12th-14th 2022, Milan, Italy. Oral presentation: *Frequency response in miRNA mediated genetic regulation (Szischik C., Ferro E., Bosia C., Ventura A.C.).*

ICSB 22: The 21st International Conference on Systems Biology. October 8th-12th 2022. Berlin, Germany. Poster presentation: *Frequency preference in miRNA-mRNA interaction: competition and cooperativity effects. Szischik C., Ferro E., Bosia C., Ventura A.C.*

XIX Congreso Regional de Física Estadística y Aplicaciones a la Materia Condensada (TREFEMAC 2022). May 11th to 13th, 2022. Universidad de La Plata, La Plata, Argentina. Poster presentation*: Estudio de redes de regulación génica con mRNA y microRNA interactuantes Szischik C., Ferro E., Bosia C., Ventura A. C.*

XLIX SAB Annual Meeting (Sociedad Argentina de Biofísica) 2021. December 1st - 3rd 2021, virtual meeting. Oral presentation of the work titled *Study of gene regulatory networks with interacting mRNAs and microRNAs, Szischik C., Ferro E., Bosia C., Ventura A.C.*

Living Light Conference 2021: *online early career researcher conference.* September 22nd -23rd 2021, virtual meeting. Oral presentation of *Electromagnetic response of corrugated multilayer systems: mirror effect in the butterfly Dione Vanillae.* Szischik C., Inchaussandague M.E.

16th Granada Seminar: *new frontiers in Nonequilibrium Statistical Physics: from fundamentals, fluctuations and hydrodynamics to biology and quantum non equilibrium* -online edition-. Instituto Carlos I de Física Teórica y Computacional, Granada, España. June 7th -17th, 2021.

105° Reunión de la Asociación Física Argentina. September 16th-30th, 2020. Virtual meeting. Oral presentation: *Estudio del efecto espejo de las alas de la mariposa Dione Vanillae.* Szischik C., Inchaussandague M.

Physical Biology of the Cell, Marine Biological Laboratory. (16 webinars) August 3rdrd to 22th 2020.

The Physics of Life Summer School 2020, Center for the Physics of Biological Function, Princeton University. June 15th to August 3rd 2020, virtual meeting.

104° Reunión de la Asociación Física Argentina. September 30th - October 3rd 2019, Santa Fe, Argentina. Oral presentation: *Análisis de opinión entorno a la interrupción voluntaria del embarazo.* Vidal M.S, Szischik C., Cicchini T. Chernomoretz A.

StatPhys27: International Conference on Statistical Physics. International Union of Pure and Applied Physics (IUPAP) July 8th to 12th, 2019, Buenos Aires, Argentina.

I Jornada de Ciencias Forenses. Facultad de Ciencias Exactas y Naturales UBA. November 1st and 2nd 2018, Buenos Aires, Argentina.

103° Reunión de la Asociación Física Argentina. September 17th to 21th 2018. Buenos Aires, Argentina. *Estudio del color de las alas de la mariposa Agraulis Vanillae: imágenes de microscopía y medidas de reflectancia.* Vidal M S, Szischik C., Inchaussandague M., Skigin D.

TEACHING EXPERIENCE

Teaching assistant in the program *Exactas Programa*. (February 2022 - March 2022) An introductory course for programming in Python for student of the Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. Selected on merits.

Teaching assistant. (July 2021 – present) Physics Department, Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. Selected on merits.

Tutor for first-year university students. (April 2021- present). Dirección de Orientación Vocacional, Secretaría de Extensión, Cultura Científica y Bienestar, Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires.

Undergraduate teaching assistant. (February 2019 - June 2021) Physics Department, Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. Selected on merits.

ADDITIONAL COURSES

Image Processing with Applications in Biomedicine. Dr. Federico Lecumberry. School of Computer Science (ECI 38), Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. 28 July – 1 August 2025. In-person course.

Bases for Inclusive Education in Higher Education, taught by the Secretariat for the Promotion of Equity and Gender, Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. May 16th to June 13rd.

Spatial and temporal correlations in experiments and simulation: a practical guide. Dr. Tomás Grigera. Course oriented to PhD. students as a complementary activity in TREFEMAC 2021 meeting (virtual). July 5th, 2021.

Writing in English course. Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. May 2021 (virtual)

Gender perspective course for teachers. Secretaría de Asuntos Académicos, Universidad de Buenos Aires. September 2020.

Data Reduction and Visualization Techniques. Dr. Daniela Rodriguez. Instituto del Cálculo. Universidad de Buenos Aires. June 2nd - August 20th, 2020. (virtual)

Science didactics seminar. Secretaría de Extensión, Cultura y Bienestar, Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. November 2019.

Quantum computing and digital physics. 33° Computer Science School. Prof. Pablo Arrighi. July 22nd-26th, 2019.

Social Data Models: Theory, Applications and critical analysis. Dr. Pablo Jensen. April 22nd-26th, 2019.

HUMAN RESOURCES TRAINING

Co-advisor in MsC./Bachelor equivalent degree thesis of Manuel Dichio. Advisor: Alejandra Ventura. 2024-ongoing. Project title: Competition regulation in cell signaling and gene expression: kinetic strategies.

Co-advisor in MsC./Bachelor equivalent degree thesis of Maia Vilcovsky. Advisor: Alejandra Ventura. Co-advised with Dr. Juan Ignacio Marrone. 2024 - ongoing. Project title: Coupled oscillators as a mechanism for achieving Temporal shifts in the implementation of biological tasks.

ACADEMIC SERVICE

Elected Graduate Representative for the Departmental Council of Physics. Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. 2024-ongoing. Participated in departmental decision-making and policy discussions.

Overseer at the PhD Subcommittee in Physics. Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires. 2023- ongoing. Participated in committee meetings evaluating PhD applicants and monitoring students' academic progress throughout the program.

OUTREARCH AND SCIENCE COMMUNICATION

Collaborator in edition Revista CHicos. Asociación Civil Ciencia Hoy. Popular science magazine dedicated to children between 9 and 13 years old. Author of the article *Detectives en acción: tensión en las sombras*, published in magazine issue 17.

Collaborator in the Statistical Physics Conference StatPhys27, organized by IUPAP. July 8th -12th, 2019. Buenos Aires, Argentina.

Collaborator in the Night of the Museums in the Facultad de Ciencias Exactas y Naturales. (2nd November 2019, 10th November 2018) Visitor guide.

Collaborator in the Physics Week in Facultad de Ciencias Exactas y Naturales. (2017-2019) Guidance and explanation of physics experiments for high school students.

TECHNICAL SKILLS

LANGUAGES

Spanish (native), English (proficient- C1, Cambridge CAE), French (basic), Italian (basic)

PROGRAMMING AND TOOLS

Python, R, MATLAB, LaTeX, Git, ImageJ

SCIENTIFIC COMPUTING

Mathematical modelling, non-linear dynamics, statistical physics, stochastic simulations, data analysis and visualization.

EXPERIMENTAL TECHNIQUES

Molecular cloning, PCR, bacterial transformation, cell culture, transfection